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fedvalidateR: R Package for Federated External Validation of Clinical Prediction Models

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Executive Summary

Clinical prediction models are increasingly evaluated across hospitals, countries and health systems (Riley et al. 2016). However, many institutions cannot transfer patient-level data because of governance, ethical, legal or operational constraints. Existing federated platforms provide important analytical capabilities, but may require dedicated client-server infrastructure or a common data model. Research networks working with heterogeneous local datasets therefore still recreate variable mapping, quality checks, validation code, output review and cross-site reporting for each study. We propose fedvalidateR, an open-source R package that will convert a multisite external-validation protocol into a reusable, version-controlled workflow. Participating sites will map local data to a shared specification, run standardised checks and analyses in ordinary R, and review a structured aggregate results object before release. Central functions will verify compatibility, compare site-specific performance, quantify heterogeneity and generate reproducible reports. The first release will support existing clinical prediction models for binary outcomes, such as mortality, clinical deterioration and hospital readmission. The project will deliver an installable R package, documented study and site-results specifications, synthetic multisite datasets, automated tests, tutorials and an end-to-end reproducible demonstration. Evaluation will use fully specified prediction models or risk scores with synthetic and publicly available data partitioned to simulate independent sites. Subject to data-access and governance approvals, the workflow may also be user-tested in Brazilian intensive care settings, building on experience from the SOFA-2 validation study, or within the ISARIC research network. The project requests US\$5,500 over 10 months, and the resulting package, documentation and reproducible examples will be made openly available to the R community.

Signatories

Project team

- **Dr Sameera Dhananjaya Viswakula**, Department of Statistics, University of Colombo — **Project Lead**: package architecture, software development, testing, documentation and release, drawing on his experience developing R packages.
- **Dr Dilanthi Priyadarshani Gamage Dona**, Department of Statistics, University of Colombo — **Co-Lead**: statistical and clinical model-validation framework, method verification, synthetic evaluation and applied user testing.
- **Research Assistant (to be appointed)**, Department of Statistics, University of Colombo — programming, synthetic-data development, testing, documentation, repository maintenance and demonstration support.

Contributors

- **Dr Leonardo dos Santos Lourenço Bastos**, Department of Industrial Engineering, Pontifical Catholic University of Rio de Janeiro, Brazil — advice on federated healthcare analytics, implementation in Brazilian intensive care settings, user testing and interpretation of findings.

The Problem

Clinical prediction models that perform well in their development data may perform differently across populations, institutions, countries and time periods because of differences in case mix, measurement practices, treatment pathways and outcome prevalence (Riley et al. 2016; Collins et al. 2015). Multisite external validation is therefore important for assessing transportability and heterogeneity in model performance.

Conventional pooled validation requires patient-level records to be transferred to a central location, which may be restricted by governance, ethical, legal or operational requirements. Federated analysis offers an alternative by retaining records locally, but participating sites must still implement consistent cohort definitions, predictor transformations, model equations, missing-data rules, performance measures and output controls. These challenges particularly affect clinical and public-health researchers, registry networks and data custodians coordinating validation across heterogeneous or resource-constrained settings.

Existing approaches address parts of this problem. DataSHIELD provides distributed analysis with client-server infrastructure and disclosure controls, including tools for assessing model performance (Gaye et al. 2014), while OHDSI PatientLevelPrediction supports reproducible network studies using data mapped to the OMOP Common Data Model (Reps et al. 2018). Other R packages provide established methods for performance estimation and evidence synthesis. However, a practical gap remains for networks whose sites can run R but cannot deploy persistent federated servers or transform entire databases to a common data model. **fedvalidateR** will address this gap by providing a lightweight, study-specific protocol-to-report workflow for local mapping, data-quality checks, model validation, reviewable aggregate outputs and cross-site synthesis. It is intended to complement, rather than replace, existing federated platforms.

The proposal

Overview

fedvalidateR will provide a standardised workflow for federated external validation of existing prediction models for binary outcomes. A coordinating team will define the model, required variables, outcome, performance measures and permitted outputs. Each institution will run the same R workflow locally to check data compatibility, calculate predicted risks and performance measures, and create a standardised results object for local review. Approved aggregate outputs can then be combined to compare performance across sites, assess heterogeneity where appropriate and generate reproducible reports.

Over 10 months, the project will develop and test the local and central workflows, synthetic multisite datasets, automated tests, documentation and an end-to-end public-data demonstration. For the R community, it will provide reusable functions, documented data structures, synthetic examples and reporting tools. It will complement rather than replace existing federated platforms and will not assume that aggregate outputs are automatically anonymous or legally compliant.

Detail

Minimum Viable Product

The minimum viable product will be a tested R package for federated external validation of fully specified prediction models and clinical risk scores for binary outcomes. It will include:

- a machine-readable study specification covering the model, predictors, outcome, prediction horizon, analyses and permitted outputs;
- tools for local variable mapping, cohort-flow documentation and data-quality assessment;
- local calculation of prespecified discrimination, calibration and overall performance measures;
- a versioned site-results object containing aggregate estimates, uncertainty measures, diagnostics, warnings and provenance information;
- central functions for compatibility checking, comparison of site-specific performance, heterogeneity assessment where appropriate and reproducible reporting; and
- synthetic multisite datasets, automated tests, tutorials and an end-to-end public-data demonstration.

The initial release will not include model development, model updating, time-to-event outcomes or multicategory outcomes; these may be considered in future versions.

Architecture

The workflow will have four components:

1. **Coordination:** create a versioned study specification defining the model, required variables, outcome, analyses and permitted outputs.
2. **Local analysis:** map a study-specific data extract to the shared specification, run data-quality checks, apply the fixed model and calculate the requested performance measures in R.
3. **Review and export:** create a structured results object containing aggregate estimates, diagnostics, warnings and metadata for local review and approval.
4. **Central synthesis:** import approved site-results objects, verify compatibility, compare performance across sites, assess heterogeneity and generate reproducible tables, figures and reports.

Patient-level data and individual predictions will remain within each institution. Approved result objects will be transferred without requiring persistent connections between sites or dedicated federated servers.

Assumptions

The project assumes that sites can run R and prepare the required local data extract; the model is fully specified and legally available for implementation; variables, outcomes and prediction horizons can be mapped consistently; sites have sufficient events and non-events for the planned analyses; and institutions can approve release of aggregate outputs.

The package will flag incompatible data or unreliable estimates rather than silently proceeding. Where site-specific mapping or data limitations prevent an analysis, the workflow will document the issue and permit compatible analyses to proceed without requiring all sites to share patient-level data.

External dependencies

The package will use a limited set of established open-source R packages, with dependencies and version requirements documented and tested. Core delivery will not depend on DataSHIELD, a common data model, server infrastructure or access to confidential data. The demonstration will use publicly available data and an openly implementable or permission-cleared prediction model; any additional real-world testing will be conditional on relevant data-access and governance approvals.

Project plan

Start-up phase

During Month 1, the Project Lead and Co-Lead will confirm responsibilities and user requirements, finalise the statistical and technical specification, agree the minimum viable product and site-results structure, and establish the public repository, open-source licence, issue tracker and automated testing framework.

Technical delivery

Period	Activities and milestones	Lead
Month 1	Finalise user requirements, specification, repository, licence and test plan.	Lead & Co-Lead
Months 2–3	Develop mapping, audit and local-validation functions; release alpha version.	Lead & RA
Months 4–5	Develop synthetic multisite datasets and site-results schema; verify statistical methods and implement automated tests.	Project team

Period	Activities and milestones	Lead
Month 6	Develop compatibility checks, central synthesis, heterogeneity assessment and reproducible reporting; release beta version.	Project team
Month 7	Complete an end-to-end demonstration using a public real-world dataset partitioned into simulated sites.	Project team
Month 8	Conduct structured user testing and, subject to approvals, explore testing in Brazilian intensive care settings or within the ISARIC research network.	Co-Lead & Contributor
Month 9	Complete independent statistical verification and code review; resolve critical issues and prepare a release candidate.	Project team & independent reviewer
Month 10	Complete tutorials, contributor documentation and governance guidance; publish and disseminate the reproducible release.	Project team

Progress will be managed through monthly meetings, repository milestones, issue tracking, code review and continuous integration.

Other aspects

Development will occur publicly on GitHub under an appropriate open-source licence, with issue tracking, contribution guidance and automated testing to support community participation. The core package will use synthetic data; any real-data testing will require relevant ethical, institutional and data-access approvals, with patient-level data remaining locally controlled. The team will disseminate major milestones and the final release through relevant R-community channels, including opportunities provided by the R Consortium. The Project Lead and Co-Lead will maintain the package after the grant, supported by automated tests, developer documentation and community contributions.

Budget & funding plan

The project requests **US\$5,500** over 10 months. Most funding supports research-assistant time for software development, synthetic-data preparation, testing, documentation and repository maintenance. Project-specific cloud services will support development and demonstration during the grant period, and user-testing support will enable structured feedback and refinement.

Budget item	Amount (US\$)	Purpose
Research Assistant	4,500	US\$450/month $\times$ 10 months for programming, synthetic-data preparation, testing, documentation and demonstration support
Project-specific cloud services	500	US\$50/month $\times$ 10 months for development, testing and demonstration
User-testing support	500	Structured testing, feedback collection and interface/documentation refinement
Total	5,500	

Funding can be released against agreed milestones, with early payments supporting specification and local-workflow development, intermediate payments supporting synthetic evaluation and central synthesis, and final payments supporting user testing, verification, documentation and public release.

Success

Definition of done

The project will be complete when a publicly available, installable **fedvalidateR** package provides the specified local-validation and central-synthesis workflows, documented study and site-results structures, automated tests, tutorials, synthetic multisite examples and an end-to-end public-data demonstration. Additional testing in Brazilian intensive care or ISARIC settings will be subject to relevant approvals and is not required for core project completion.

Measuring success

Success will be assessed by:

- passing R package checks and all critical automated tests;
- agreement with independent reference calculations within prespecified tolerances;
- successful end-to-end analysis of at least three contrasting synthetic sites;
- at least **80% test coverage** for core functions;
- reproducible tutorials and demonstration materials;
- structured feedback from at least **five intended users**, with critical issues resolved; and
- public release of the package, code and documentation.

Future work

Future releases may extend **fedvalidateR** to other outcome types, model recalibration, decision-curve analysis, additional disclosure controls and DataSHIELD integration. Further development may also include multilingual documentation and CRAN submission, guided by user and community feedback.

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